Workshop 2 – Kaggle

Mechanisms of Action (MoA) Prediction – Kaggle Competition

Course: Systems Analysis

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1 Analysis and Requirements

1.1 Context and problems identified

Predicting the Mechanisms of Action (MoA) of chemical compounds is a central challenge in bioinformatics and drug discovery, as it allows anticipating cellular effects before conducting costly laboratory experiments. The Kaggle MoA prediction competition addresses this challenge through a multi-label machine learning approach, where each sample can be associated with multiple MoAs simultaneously.

The analysis reveals that the system faces several critical issues that affect its performance and reliability:

- **High data dimensionality:** The dataset includes over 800 gene expression variables (g-variables) and about 100 cellular feature variables (c-variables). This complexity increases the likelihood of indirect relationships and structural correlations, which can amplify errors if not properly managed.

- **Non-linear interactions:** The relationships between genes, cellular features, and experimental metadata (`cp_type`, `cp_dose`, `cp_time`) are non-linear, meaning small variations can lead to disproportionate changes in MoA predictions.

- **Class imbalance:** The presence of rare MoAs makes it difficult for the model to generalize and makes the training/validation split particularly sensitive.

- **Inherent randomness in models:** Neural networks and ensemble methods introduce variability due to random parameter initialization and training subset selection, leading to different results even with the same input data.

- **Critical dependencies among inputs:** Variables such as treatment type (`cp_type`) determine which samples are evaluable and modulate prediction sensitivity. Likewise, correlated g- and c-variables produce complex interactions that require careful management to avoid error amplification.

These challenges not only represent technical issues but also highlight the need for a robust system design that accounts for the sensitivity and chaotic nature of the problem, ensuring reproducible and reliable results.

1.2 Data Analysis and Relationships Between Elements

The system consists of three main types of inputs:

- **Biological data:**
  - **g-variables:** Expression levels of 772 genes representing the molecular response of cells.
  - **c-variables:** Cellular features under different experimental conditions (approximately 100 variables).
- **Experimental metadata:**
  - `cp_type`: Indicates whether the sample is a control or treated.
  - `cp_dose`: Indicates the dose applied.
  - `cp_time`: Duration of treatment.

The analysis reveals complex and dynamic relationships among these variables:

- The activation of certain genes induces changes in cellular characteristics, generating cascading effects.
- Experimental metadata modulates the intensity and timing of the cellular response, directly affecting the likelihood of MoA activation.

During preprocessing, normalization, dimensionality reduction, and feature selection are applied to reduce excessive correlations and ensure that critical interactions are properly captured by the models.

This flow of information shows that the system is highly interdependent and non-linear, meaning that any alteration in input data or intermediate processes (e.g., preprocessing or feature selection) can significantly affect the final outcomes.

1.3 Sensitivity and Chaos

The analysis of sensitivity and chaotic aspects of the system reveals the following:

- **Highly sensitive parameters:** `cp_type`, `cp_dose`, `cp_time`, g-variables, and c-variables. Even small changes in these inputs can significantly alter the probability of multiple MoAs.
- **Non-linear effects and feedback:** Multi-label model training occurs in iterative cycles (normalization $\rightarrow$ modeling $\rightarrow$ evaluation $\rightarrow$ hyperparameter tuning), where each iteration feeds into the next. This enhances robustness but also increases the system's sensitivity to small changes.

- **Sources of chaos:** Random initialization of neural networks, processing order in ensemble models, and the way data is split (train/validation) introduce uncertainty into the results, especially for underrepresented MoAs.
- **Sensitivity control:** Techniques such as drug-aware splitting, variable normalization, and selection of critical features help reduce unpredictable effects and ensure the model generalizes correctly.

1.4 Functional Requirements

Based on the findings from the analysis, the system must fulfill the following functions:

Data ingestion and preprocessing:

- Handle high-dimensional tabular data, including g-variables, c-variables, and experimental metadata.
- Apply normalization, dimensionality reduction, and outlier correction to prepare the data for modeling.

Multi-label modeling:

- Train models that capture non-linear interactions and relationships between biological variables and metadata.
- Generate probability vectors for each MoA, maintaining internal consistency and sensitivity of the system.

Evaluation and feedback:

- Compare predictions with known outcomes using standardized metrics (e.g., log loss).
- Adjust models and preprocessing iteratively to improve accuracy and stability.

1.5 Non-Functional Requirements

The system must also meet quality criteria to ensure its reliability and scalability:

- **Performance:** Efficient processing despite high dimensionality and dataset size.
- **Reliability:** Reproducible results in the face of random variations during training and validation.

- **Scalability:** Ability to incorporate new data or experiments without modifying the core architecture.
- **Maintainability:** Clear documentation, version control for data and models, and traceability of results.
- **Interpretability:** Outputs must be understandable by analysts and biologists to support decision-making in biomedical research.

2 General System Architecture

2.1 Overview

The system has been designed for the accurate prediction of the Mechanism of Action (MoA) of pharmacological compounds, using a foundation of genetic and cellular datasets. Its structure is organized into well-defined modules that ensure efficient workflow management, from the reception of initial data to the generation of predictive outputs.

2.2 Main System Modules

1. **Data Ingestion:** Responsible for loading and verifying input files (training, test, metadata). Ensures data integrity, completeness, and formatting.
2. **Processing:** Involves cleaning outliers, normalizing variables, and handling parameters such as `cp_type`, crucial for valid sample selection.
3. **Model Construction:** Core phase where Machine Learning algorithms are selected and trained, from linear to deep neural networks.
4. **Evaluation and Validation:** Measures predictive performance using statistical metrics and cross-validation to ensure model stability.
5. **Result Generation:** Produces final predictions and formats them into the required standard output file (e.g., `submission.csv`).

2.3 Operational Workflow of the System

1. **Input:** Reception of high-dimensional biological data.
2. **Transformation:** Data cleaning, scaling, and preparation.

3. **Training:** Model learns and assimilates patterns associated with mechanisms of action.

4. **Evaluation:** Performance assessment and hyperparameter tuning.

5. **Output:** Final predictions ready for competition or real use.

2.4 Technical Components and Organization

- **Base Language:** Python (.py)

- **Development Environment:** Notebooks and specialized experimental environments.

- **Modular Structure:** Independent phases allow updates (e.g., replace modeling algorithm without affecting preprocessing).

- **Scalable Approach:** Architecture supports integration of Deep Learning techniques or automated pipelines.

2.5 Purpose of the Architecture

The main goal is to maintain a structured, traceable, and reproducible system capable of handling complex biological datasets while remaining flexible to future requirements and predictive model enhancements.

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3 Chaos and Sensitivity

3.1 Characterization of Sensitive Data in the Context of the MoA System

Within this system, data sensitivity manifests in two primary dimensions:

- **Biomedical Data of Cellular and Genetic Origin:** The datasets originate from high-throughput experiments with human cell lines. While they do not constitute "personal data" in the traditional sense (such as a name or national ID), genetic and protein expression information is inherently sensitive. There is a theoretical risk, albeit remote in this processed context, that it could be used to infer information about populations or even, in future scenarios, be linked to individuals.

- **Intellectual Property and Competitive Advantage:** The data and the compounds under investigation are the property of the sponsoring entities. The premature or unauthorized disclosure of chemical structures, mechanisms of action, or discovered compound-gene relationships could compromise patents, R&D strategies, and market positioning. This dimension turns the data into a confidential commercial asset.

Systemic Architecture and Data Control Mechanisms

The system implements a layered architecture with specific controls to mitigate the risks associated with sensitive data.

Layer 1: Preprocessing and Anonymization at the Source

Before being ingested into the central system, the data undergoes rigorous preprocessing by the providers. This process includes:

- **Aggregation and De-identification:** Gene expression and cellular data are presented in an aggregated form, removing any direct identifiers linked to specific individuals.

- **Normalization and Encoding:** Features are normalized and assigned generic codes (e.g., g-1, g-2 for genes; c-1, c-2 for cellular data). This eliminates actual gene names and obfuscates direct biological interpretation without an internal dictionary, protecting sensitive contextual information.

Layer 2: Infrastructure and Access Control in the Central Platform

The central platform acts as the execution and collaboration environment, implementing the following safeguards:

- **Offline Execution Environment:** For the inference and prediction submission phases, external users do not download the test set. Instead, they execute their models within an isolated and controlled environment inside the platform, where the system directly accesses the test data. This prevents the exfiltration of the evaluation set.

- **Digital Non-Disclosure Agreement (NDA):** All users must accept the terms of use, which function as a legal agreement prohibiting them from sharing, downloading, or attempting to reconstruct the test datasets. This is a critical contractual barrier.

- **Rate Limiting:** The system imposes a limit on the number of daily submissions. This not only prevents statistical overfitting on the test set but also hinders inference attacks that seek to reconstruct data through multiple queries to the evaluation system.

Layer 3: Post-Processing and Dissemination of Results

Upon process completion, the system manages the output of information:

- **Publication of Methods, Not Raw Data:** The publication of model source codes can be encouraged, which promotes scientific transparency. However, the sensitive test data on which these models were evaluated is not released to the public.

- **Aggregated Results:** The system's primary output is the MoA predictions for the test compounds, which are aggregated and anonymized. The focus is on the predictive outcome rather than the complete exposure of the underlying input data.

Risk Assessment and Ethical Considerations

Despite the robust controls, areas of risk can be identified:

- **Inference Model Attacks:** A malicious user could design a model that, in addition to predicting the MoA, attempts to infer and reconstruct specific features of the test input data from interactions with the evaluation system.

- **Information Leakage through Metadata:** An exhaustive analysis of public training data, combined with domain knowledge, could allow for the partial deduction of compound-gene relationships that are confidential.

The system mitigates these risks through its rule design (rate limiting, NDA) and the very nature of the task, which is an extremely complex multivariable prediction, making inference attacks computationally unfeasible and impractical.

The Mechanisms of Action (MoA) prediction system operates under a paradigm of minimum privilege and maximum containment, which implies that external users receive an anonymized training set and work on a test set to which they never have direct access, within an isolated computational environment.

The strategy effectively combines technical anonymization at the source, contractual and platform access controls, and an architecture that clearly separates model development from final evaluation. This comprehensive approach allows the system to fulfill its primary objective while protecting sensitive biomedical data assets, establishing a robust standard for handling critical information in the fields of bioinformatics and pharmaceutical development.

3.2 Sources of Instability and Critical Sensitivity

The MoA system, due to its high-dimensional and complex nature, presents several strange attractors where small perturbations can lead to significant and potentially chaotic variations in its results and operation.

Sensitivity in the Data Layer: The Butterfly Effect in Preprocessing

The initial phase of anonymization and encoding, while crucial for protection, introduces a point of high sensitivity.

- **Perturbation:** A minimal variation in the encoding protocol (for example, a different random seed for normalization, a slightly different criterion for noise filtering) between two batches of data.
- **Chaotic Amplification:** This small discrepancy leads to a slightly different feature space. Given that machine learning models (especially deep neural networks) are highly sensitive to these geometries, the same model architecture trained on each batch of data can converge towards fundamentally different solutions. A gene coded as g-78 in one batch and with a slight scaling variation in another can completely alter the importance a model assigns to it, changing the inferred dependency network and, ultimately, the final MoA prediction. This hinders reproducibility and the consolidation of results.

Sensitivity in the Modeling Layer

The predictive core of the system is inherently a chaotic system in the parameter space.

- **Perturbation:** The random initialization of weights in a neural network model, an infinitesimal variation in the learning rate, or the order of presentation of training data.
- **Chaotic Amplification:** In a non-convex optimization space with billions of parameters, these small initial differences can lead the training process towards radically different local minima. Two models with identical architecture, faced with the same task, can find solutions with similar accuracies but which are based on completely different internal inference mechanisms. This lack of stable convergence towards a unique solution represents a chaotic variation in the system's "explainability," where the prediction is correct, but the underlying reason is unpredictable.

Sensitivity in the Evaluation Layer: The Instability of the Isolated Environment

The evaluation mechanism in the isolated environment, designed for security, creates a point of critical sensitivity for feedback.

- **Perturbation:** A minor, almost imperceptible alteration in the distribution of the evaluation test set (for example, an inadvertent selection bias in the compounds chosen for the final test).

- **Chaotic Amplification:** Since external users adjust their models exclusively in response to the performance metric reported by this single test set, a minimal bias in this set can direct the entire development community towards optimizing for a distorted reality. This generates an "arms race" where models become extremely adept at predicting the biased test set, but their generalization and utility in the real world is compromised. The system as a whole converges towards a locally optimal but globally inferior solution.

Sensitivity in the Human-System Interface: The Interpretation of Results

The output of the system is a point of high sensitivity for decision-making.

- **Perturbation:** A small variation in the confidence score of a prediction (e.g., a compound moves from an 89% to a 91% probability for a specific MoA).
- **Chaotic Amplification:** In the context of drug development, where decisions involve investments of millions and human lives, this minimal numerical difference can drastically alter the research trajectory. A team may decide to discard a promising compound or, conversely, invest significant resources in a false positive, based on a fluctuation that may be an artifact of the model's noise or internal chaotic sensitivity. The quantitative output of the system is interpreted in a qualitative and binary manner, disproportionately amplifying the importance of small variations.

The MoA prediction system, far from being a deterministic and stable mechanism, operates in a high-sensitivity regime where small variations in preprocessing, modeling, evaluation, and interpretation can be amplified chaotically, leading to disparate results and unpredictable critical decisions. The stability of the system is not achieved by eliminating these sensitivities (which is very difficult due to its complexity) but through robust governance that recognizes and mitigates them.

This implies the implementation of strict reproducibility protocols (fixed seeds, versioned preprocessing pipelines), continuous evaluation on multiple validation sets to avoid overfitting to a single sensitive evaluation point, and, crucially, the interpretation of the model within a framework of uncertainty, where predictions are understood as unstable probabilities in a chaotic space and not as deterministic truths.

3.3 Sources and Classification of Uncertainty in the MoA System

Epistemic Uncertainty

Arises from the system's lack of knowledge or information.

- **Limited and Biased Data:** The chemical and biological space is virtually infinite, while the training dataset is finite and potentially biased. The system lacks information about structurally novel compounds or mechanisms of action not represented in the training data. This uncertainty manifests as low confidence in predictions for molecules that fall outside the distribution of the training data.
- **Model Complexity and Overfitting:** Complex models, such as deep neural networks, can learn spurious patterns or noise in the training data instead of the underlying biological signal. This generates structural uncertainty: the model is "confident" in its predictions for the training data but highly uncertain and erroneous when faced with new data, as its knowledge does not generalize.
- **Hidden or Unmeasured Features:** The system operates on a predefined set of features (gene expression, cell viability). There is a fundamental uncertainty derived from omitted variables or biological mechanisms that are not captured by these measurements but are crucial for determining the MoA.

Aleatory Uncertainty

Is inherent to the randomness of the biological process itself and cannot be eliminated, only characterized.

- **Biological Experimental Noise:** High-throughput screening experiments are subject to inherent experimental variability. The gene expression of a cell line can fluctuate due to uncontrolled environmental factors, introducing stochastic noise into the input data that the system cannot distinguish from the real signal induced by the compound.
- **Stochasticity in Biological Mechanisms:** The drug-biology interaction is not completely deterministic. The same compound can exert slightly different effects between experimental replicates or cell populations due to the probabilistic nature of biochemical pathways. This variability is directly incorporated into the system's uncertainty.

Propagation and Amplification of Uncertainty

Uncertainty does not remain isolated; it propagates and amplifies through the system's layers:

1. **Data Layer:** Experimental noise introduces uncertainty into the raw data.
2. **Modeling Layer:** The machine learning model absorbs this uncertainty and combines it with its own epistemic uncertainty. During training, the model learns to map noisy inputs to outputs, but the function it learns is inherently uncertain.

3. **Output Layer:** The uncertainty condenses into the final prediction. A well-designed system would not only output a MoA label but also a confidence metric. However, if the uncertainty is not quantified correctly in the previous layers, the reported confidence will be misleading.

Consequences of Unmanaged Uncertainty

The lack of rigorous uncertainty quantification carries critical operational and scientific risks:

- **False Certainty:** The most severe risk is the system presenting a prediction with high confidence that is incorrect. This can lead to allocating resources to ineffective compounds or discarding promising candidates, resulting in significant economic losses and wasted time in drug development.

- **Suboptimal Decisions:** Without a clear understanding of the system's knowledge limits, decision-making is based on incomplete information, increasing the likelihood of failures in later clinical stages, where costs multiply.

Error Monitoring in the MoA Prediction System

The system implements a stratified monitoring framework that spans from technical errors to scientific discrepancies:

Technical-Operational Monitoring

- **Input Data Validation:** Verification of formats, valid ranges, and consistency of genomic and cellular data.

- **Performance Metrics Tracking:** Continuous monitoring of loss during training and validation metrics (accuracy, F1-score).

- **Data Drift Detection:** Systems that alert about changes in input data distribution compared to the original training set.

Scientific-Interpretive Monitoring

- **Systemic Error Analysis:** Identification of patterns in false positives/negatives according to compound classes or mechanisms.

- **External Cross-Check Validation:** Periodic comparison of predictions with existing scientific literature and pharmacological databases.

- **Calibration Monitoring:** Verification that prediction probabilities correspond to actual accuracy rates.

Response Mechanisms

- **Escalation System:** Classification of errors by criticality (technical, interpretive, safety-related).
- **Feedback Loop:** Protocols to reintegrate error findings into the model retraining process.
- **Decision Audit:** Recording of predictions with high uncertainty for subsequent experimental validation.

Error Governance

- **Centralized Dashboard:** Unified visualization of system status and critical error metrics.
- **Automated Alerts:** Proactive notifications of deviations from expected error patterns.
- **Periodic Reviews:** Scheduled analysis of false positives/negatives to refine acceptance criteria.

3.4 Dimensions of Model Stability

Stability must be evaluated across multiple interdependent dimensions:

Computational Stability

Refers to the consistency of the model's results across different runs of the training process.

- **Source of Instability:** The random initialization of weights, the order in which data is presented, and the stochastic nature of optimizers can cause the same model, with the same architecture and data, to converge to different solutions in each training session.
- **Impact:** This instability manifests as significant variation in performance metrics between different "replicas" of the model, undermining reproducibility and confidence in any single instance of the model.

Data Stability

Evaluates the model's sensitivity to small perturbations in the input data, whether due to experimental noise or slight variations in distribution.

- **Source of Instability:** The inherent noise in high-throughput biological experiments, batch variations in reagents, or technical differences between laboratories can introduce variations in the input feature vectors.
- **Impact:** An unstable model will significantly alter its predictions in response to these minimal perturbations. A compound predicted to have MoA "A" could be classified under MoA "B" after a slight fluctuation in a gene's reading, which is unacceptable for a pharmaceutical application.

Structural Stability

Defines the model's ability to maintain its performance when faced with data that comes from a different distribution than the training data (Out-of-Distribution).

- **Source of Instability:** The training set covers only a finite subset of the chemical and biological space. When the system is used to predict the MoA of structurally novel compounds or those without analogous data in the training set, its behavior becomes unpredictable.
- **Impact:** The model can suffer a drastic drop in performance or, worse, generate predictions with high confidence that are incorrect.

Threats to Stability in the MoA System

The system architecture introduces critical points that threaten stability:

- **High Dimensionality:** The feature space is extremely vast compared to the number of training samples. This creates an environment prone to overfitting, where the model memorizes noise instead of learning the underlying biological signal, resulting in a complex but unstable model.
- **Spurious Correlations:** The model may stabilize based on correlations that are not causally related to the MoA. When this artifact is not present in the production data, the model becomes destabilized.
- **Feature Interdependence:** Genetic and cellular data are highly correlated. A small perturbation can propagate through these dependency networks, amplifying its effect and destabilizing the final prediction.

Strategies to Ensure Model Stability

A robust system must integrate specific techniques to mitigate threats to stability:

Modeling and Training Level Techniques

- **Advanced Regularization:** Use of techniques to penalize model complexity and prevent overfitting, thus fostering simpler and more stable solutions.
- **Ensemble Models:** Combining predictions from multiple models averages out individual instabilities. An ensemble is inherently more stable and robust than a single model, as it smooths out variations from different initializations or data subsets.
- **Extensive Cross-Validation:** Used not only for hyperparameter selection but also to assess the stability of the model's performance across different data partitions. Low variance in cross-validation metrics is a key indicator of stability.

Data and Preprocessing Level Techniques

- **Robust Feature Engineering:** Dimensionality reduction using techniques to project the data into a denser, more representative latent space, removing noise and redundancy.
- **Data Augmentation:** Introducing controlled and realistic variations into the training data to force the model to learn features invariant to these perturbations, thereby improving its robustness.

System Level Techniques

- **Continuous Drift Monitoring:** Implementing a system that continuously monitors the distribution of input data in production and compares it to the training distribution. Detecting "data drift" or "concept drift" serves as an early warning of potential model destabilization.
- **Automated and Controlled Retraining Pipelines:** Establishing clear protocols for retraining the model with new data, ensuring that each new version is rigorously evaluated for stability and performance before deployment in a production environment.

4 Tool Selection

For the implementation of the Mechanism of Action (MoA) prediction system, tools were selected that enable efficient handling of high-dimensional data, training of multi-label machine learning models, and clear, reproducible visualization of results. The decisions were based on criteria of robustness, scalability, maintainability, and alignment with best practices in systems engineering.

Programming language: Python was chosen as the main programming language due to its widespread adoption in bioinformatics and data science, compatibility with advanced machine learning libraries, and ease of integrating data processing, modeling, and evaluation workflows.

Data preprocessing and manipulation: **Pandas** and **NumPy** will be used to perform data cleaning, handle missing values, normalization, and dimensionality reduction—critical tasks to ensure the model receives consistent and high-quality information.

Modeling and machine learning: For model construction, **scikit-learn** will be used for traditional multi-label classification techniques, and **TensorFlow** or **PyTorch** for developing neural networks and more complex ensemble models. These libraries support control of initialization randomness, cross-validation, and efficient hyperparameter tuning.

Visualization and result analysis: **Matplotlib**, **Seaborn**, and **Plotly** will be used to generate correlation plots, distributions, and pattern visualizations in the data, facilitating the interpretation of results and identification of critical variables influencing predictions.

Version control and collaboration: The project will be managed using **GitHub**, enabling version tracking, change traceability, and facilitating collaboration among team members.

Justification: The combination of these tools ensures that the system is robust, scalable, and reproducible, adhering to systems engineering principles and ensuring that the predictions generated are reliable, interpretable, and useful for bioinformatics research.

5 Implementation Scheme

The implementation of the Mechanism of Action (MoA) Prediction System follows a modular and reproducible workflow divided into five main stages. Each stage has a specific goal, ensuring data integrity, model robustness, and secure deployment.

1. **Data Acquisition and Preprocessing**
   - Collect the training, test, and metadata files from the competition dataset or internal repository.
   - Perform cleaning, normalization, and feature encoding using Python scripts (`pandas`, `NumPy`).
   - Apply anonymization and validation protocols before integration into the main pipeline.
2. **Feature Engineering and Selection**
   - Apply dimensionality reduction (e.g., PCA or autoencoders) to minimize noise.
   - Encode categorical variables and generate new derived features relevant to the MoA prediction task.
3. **Model Training and Validation**
   - Train models using machine learning or deep learning frameworks (`scikit-learn`, `TensorFlow`, `PyTorch`).
   - Use stratified cross-validation to ensure model generalization and robustness.
   - Log metrics such as accuracy, F1-score, and loss evolution for continuous monitoring.
4. **Evaluation and Prediction Generation**
   - Evaluate the model on the isolated test set within a controlled environment.
   - Generate the final prediction file (`submission.csv`) with MoA probabilities for each compound.
   - Assess model stability and uncertainty before submission.
5. **Deployment and Documentation**
   - Maintain all source code under version control using GitHub.
   - Compile the final report and documentation in LaTeX, generating both `Workshop2.pdf` and `README.md`.

Implementation Workflow Summary

Input → Data Collection and Cleaning
Process → Feature Engineering, Model Training, Evaluation
Output → MoA Predictions, Stability Analysis, Final Report

5.1 Modular Prediction System for Mechanisms of Action (MoA)

The prediction system for Mechanisms of Action (MoA) is organized into a modular workflow that ensures data integrity, model robustness, and prediction reliability. Each stage fulfills a specific role within the information processing pipeline, without focusing on specific tools or libraries.

Data Ingestion and Preprocessing Experimental data are received, including gene expression levels, cellular features, and experimental metadata.

Data cleaning, normalization, and control of experimental variability are performed to ensure the information is consistent and suitable for modeling.

This stage allows for the detection and mitigation of issues related to high dimensionality, excessive correlations, and missing values.

Feature Engineering and Selection The most relevant variables for MoA prediction are identified, reducing noise and redundancy in the data.

The data are transformed in such a way that critical relationships among genes, cellular features, and experimental metadata are captured by the model.

This stage ensures that the most sensitive and decisive information is prioritized.

Modeling and Training Supervised multi-label learning models are applied to capture nonlinear interactions and dependencies among variables.

Training and validation iterations are performed to tune model parameters and ensure generalization capability.

This stage reflects the system's sensitivity: small variations in the data can alter the predicted probabilities of multiple MoAs simultaneously.

Evaluation and Prediction Generation Model predictions are compared with known data using standardized metrics that reflect accuracy and reliability.

A conceptual report of MoA activation probabilities is generated, enabling the identification of system stability and reproducibility.

This stage closes the feedback loop, where results guide potential adjustments in preprocessing and feature selection.

System Control and Traceability The entire workflow is designed to ensure that changes in data or processes can be traced and evaluated.

Each module is dependent on the preceding one, minimizing error propagation and ensuring reproducibility of results.

A conceptual framework for documentation and tracking of each system stage is established, ensuring maintainability and scalability.

6 Conclusions

- The management of sensitive data in the MoA prediction system is supported by a multi-layered strategy that prioritizes containment over exposure. Through technical anonymization at the source, restricted access to isolated environments, and contractual controls, the system achieves an effective balance between open scientific collaboration and the protection of intellectual property and biomedical privacy, transforming critical data into safely usable information. The proactive management of inherent uncertainty completes this framework, ensuring that confidentiality does not compromise the reliability of predictions.

- The developed architecture demonstrates that the prediction of mechanisms of action does not rely solely on the model, but on the rigorous coordination of the entire system. From data ingestion to the generation of the final output file, each module plays a critical role in preserving coherence, traceability, and reproducibility. Proper flow management—particularly during preprocessing and validation stages—ensures that the model receives consistent data and that the resulting predictions can be interpreted with reliability.

7 References

References

[1] Kaggle. *Mechanism of Action (MoA) Prediction Competition*. Kaggle, 2020. `https://www.kaggle.com/c/lish-moa`.